SE 226 – LAB#7 2025-2026 SPRING

Aim: File Handling, Classes and Inheritance.

In today's lab, you will build a fleet management system that can handle different types of vehicles (Cars, Trucks, and Motorcycles). You will use class inheritance to handle different vehicle types, and file handling to save and load these vehicles from a text file.

1. Create a base class Vehicle with three attributes:
   - vid (string - Vehicle ID)
   - model (string)
   - year (integer)

   Also define:
   - A constructor method __init__ to initialize these attributes.
   - A __str__ method to display items in a readable format.
   - A __eq__ method that compares two vehicles based on their vid.
   - A method called is_new(n) that returns True if the vehicle's year is within the last n years from 2026.

2. Create three subclasses of Vehicle:
   - **Car:** adds two attributes: fuel_type (e.g., Electric, Hybrid, Petrol) and doors (integer).
   - **Truck:** adds two attributes: max_load (integer, in kg) and axles (integer).
   - **Motorcycle:** adds two attributes: engine_cc (integer) and type (e.g., Sport, Cruiser).

   Each subclass must override the __str__() method to include all attributes in a human-readable format.

3. Create a function called save_fleet_to_file(vehicles, filename):
   - This function takes a list of vehicle objects and a file name.
   - Each object should be saved on one line, formatted manually based on its type, e.g.:
     i. Car, V001, Tesla Model 3, 2023, Electric, 4

4. Create a function called load_fleet_from_file(filename):
   - This function reads each line of the file.
   - Based on the first item on each line (e.g., Car, Truck, Motorcycle), create the correct object manually using if-else statements.
   - Return the list of reconstructed objects.

5. In your script, create at least 6 vehicle items:
   - At least 2 of each type: Car, Truck, Motorcycle.
   - Save them to a file named fleet.txt using save_fleet_to_file().
   - Load the items back using load_fleet_from_file().
   - Print each item using a loop.

6. Apply filtering:
    - Print only the vehicles added in the last 4 years (using is_new(4)).
    - Print only the Car objects where the fuel type is "Electric".

SAMPLE OUTPUT

```
Loading fleet data from 'fleet.txt'...
  6 vehicles loaded successfully.

  --- All Vehicles ---
  [Car]        VID: V001 | Tesla Model 3   (2023) | Fuel: Electric | 4 Doors
  [Truck]      VID: T101 | Volvo FH16      (2019) | Load: 25000kg  | 6 Axles
  [Motorcycle] VID: M301 | Yamaha R1       (2024) | Eng: 998cc     | Type: Sport
  [Car]        VID: V002 | Toyota Corolla  (2018) | Fuel: Petrol   | 4 Doors
  [Truck]      VID: T102 | Mercedes Actros (2021) | Load: 18000kg  | 4 Axles
  [Motorcycle] VID: M302 | Harley Davidson (2015) | Eng: 1200cc    | Type: Cruiser

  --- Recent Vehicles (Last 4 Years) ---
  [Car]        VID: V001 | Tesla Model 3   (2023) | Fuel: Electric | 4 Doors
  [Truck]      VID: T102 | Mercedes Actros (2021) | Load: 18000kg  | 4 Axles
  [Motorcycle] VID: M301 | Yamaha R1       (2024) | Eng: 998cc     | Type: Sport

  --- Electric Cars Only ---
  [Car]        VID: V001 | Tesla Model 3   (2023) | Fuel: Electric | 4 Doors
```